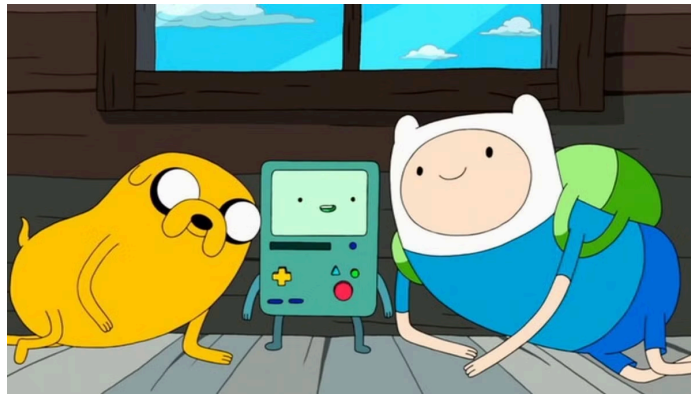


W11 Exercises

Task 0

- Download the dataset `adventure_log`

Task 1: Adventure Time!



💬 Finn the Human and Jake the Dog travel across the Land of Ooo, discovering mysterious places, meeting unusual creatures, and experiencing countless adventures

- During their journeys, they keep a log `adventure_log` of interesting events they encounter. The log includes the following columns:
 - `name` : the character(s) involved in the adventure
 - `start_str` : the starting date of the adventure
 - `end_str` : the ending date of the adventure
 - `location` : the place where the adventure happened
 - `event` : the type of adventure or discovery



However, since Finn and Jake are both very scatterbrained, the adventure dates were recorded in different formats, and some records are incomplete

- Follow the following instructions to help Finn and Jake organize the adventure log and analyze their adventures during Oct. 2026
 - Import `dplyr`
 - Use the pipe operator `%>%` throughout the analysis

Step 1: Clean the Adventure Dates

- Create a new dataframe `log_complete` and add `start_date` and `end_date` by converting `start_str` and `end_str` into date format
 - **Hint:** keep missing values as `NA` using `mutate()`
 - **Hint:** use `grepl("-", start_str, ) ~` in `case_when()`

Step 2: Remove Incomplete Adventure Records

- Remove all rows containing missing values

Step 3: Expand Adventures into Daily Records

- Create a new dataframe `adventure_coverage`. Each adventure should become one row per day (e.g., if the adventure is from `2026-10-01` to `2026-10-03`, then there will be 4 rows that take values `2026-10-01`, `2026-10-02`, `2026-10-03` in date column)
 - **Hint:** for each row in `log_complete`, create a temporary dataframe `temp` in the `for` loop with columns `name`, `location`, `event`, and `date`, where `date` columns includes all the dates between `start_date` and `end_date`
 - **Hint:** use `rbind()` to combine dataframes in each loop

Step 4: Month Information

- Create a new column `month` containing only the month information (e.g., `2026-10-01` → `10`)
- Create a new dataframe `log_oct` containing only adventures in Oct. 2026

Step 5: Analyze October Adventures

- **Q1:** How many days did each character spend on adventures in Oct. 2026?
 - **Hint:** use `distinct()` to find all characters' name
 - **Hint:** `Finn & Jake` are 2 characters but recorded in one entry
 - We will learn a better way to answer this question with `distinct()` and `summarize()` next week
- **Q2:** Which character had the most adventure days in October?